G11Adv	Mathematics	الرياضيات	T1 · Practice

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Answer key · solutions	مفتاح الإجابة والحلول

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Answer key · solutions

الحلول

Answer key · solutions — **مفتاح الإجابة والحلول**

1 · Practice · Ms. Fatima Al Saeedi · slide 7 · answer B

For $f(x) = 3^x$, which statement is true?

1. $f(0) = 3^0 = 1 \Rightarrow (0, 1)$ put x = 0
2. $\boxed{3^x > 0 \text{ for every } x \Rightarrow y = 0}$ the graph never touches y = 0

2 · Practice · Ms. Fatima Al Saeedi · slide 7 · answer B

For $f(x) = 3^x$, what happens to $f(x)$ as $x \to -\infty$ and as $x \to \infty$?

1. $3 > 1 \Rightarrow \text{growth}$ base bigger than 1
2. $\boxed{x \to -\infty : \ 3^x \to 0 \quad x \to \infty : \ 3^x \to \infty}$

3 · Practice · Ms. Fatima Al Saeedi · slide 13 · answer C

Which description of the transformation matches $f(x) = -2(4)^x$?

1. $f(x) = a\,b^x, \ a = -2$
2. $|a| = 2 > 1 \Rightarrow \text{vertical stretch}$
3. $\boxed{a < 0 \Rightarrow \text{reflection on the } x\text{-axis}}$

Answer key · solutions	الحلول
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4 · Practice · Ms. Fatima Al Saeedi · slide 34 · answer A

Bacteria A: $y = 20e^{0.079t}$. Bacteria B: $y = 52e^{0.064t}$. After how many minutes are there more A cells than B cells?

1. $20e^{0.079t} > 52e^{0.064t}$
2. $e^{0.015t} > 2.6$ divide by 20 and by e^{0.064t}
3. $\boxed{t > \dfrac{\ln 2.6}{0.015} \approx 63.7}$

5 · Practice · Ms. Fatima Al Saeedi · slide 27 · answer D

Write the equation in exponential form. $\log_{15} 225 = 2$

1. $\log_b x = a \Leftrightarrow b^a = x$ the base stays the base
2. $\boxed{\log_{15} 225 = 2 \Rightarrow 15^2 = 225}$

6 · Practice · Ms. Fatima Al Saeedi · slide 27 · answer A

Write the equation in exponential form. $\log_5 \dfrac{1}{25} = -2$

1. $\log_b x = a \Leftrightarrow b^a = x$
2. $\boxed{5^{-2} = \dfrac{1}{25}}$

Answer key · solutions	الحلول
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7 · Practice · Ms. Fatima Al Saeedi · slide 27 · answer A

Write the equation in exponential form. $\log_3 \frac{1}{27} = -3$

1. $\log_b x = a \Leftrightarrow b^a = x$
2. $\boxed{3^{-3} = \frac{1}{27}}$ base 3, exponent −3

8 · Practice · Ms. Fatima Al Saeedi · slide 27 · answer B

Write the equation in exponential form. $\log_4 64 = 3$

1. $\log_b x = a \Leftrightarrow b^a = x$
2. $\boxed{4^3 = 64}$

9 · Practice · Ms. Fatima Al Saeedi · slide 103 · answer A

Solve the equation. $\log_4(2x^2 - 4) = \log_4(2x)$

1. equal bases ⇒ equal arguments
2. $2x^2 - 4 = 2x \Rightarrow x^2 - x - 2 = 0 \Rightarrow x = 2$ or $x = -1$; $x = -1$ gives $2x < 0$ reject the value that makes a log argument negative
3. $\boxed{x = 2}$

Answer key · solutions	الحلول
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10 · Practice · Ms. Fatima Al Saeedi · slide 103 · answer A

Solve the equation. $\log_5(x^2-6)=\log_5(x)$

1. equal bases $\Rightarrow$ equal arguments
2. $x^2-6=x \Rightarrow x^2-x-6=0 \Rightarrow x=3$ or $x=-2$; $x=-2$ gives $x<0$ reject the value that makes a log argument negative
3. $\boxed{x=3}$

11 · Practice · Ms. Fatima Al Saeedi · slide 103 · answer A

Solve the equation. $\log_3(x^2-8)=\log_3(2x)$

1. equal bases $\Rightarrow$ equal arguments
2. $x^2-8=2x \Rightarrow x^2-2x-8=0 \Rightarrow x=4$ or $x=-2$; $x=-2$ gives $2x<0$ reject the value that makes a log argument negative
3. $\boxed{x=4}$

12 · Practice · Ms. Fatima Al Saeedi · slide 29 · answer A

Express in terms of common logarithms, then approximate to the nearest ten-thousandth. $\log_2 11$

1. $\log_2 11=\dfrac{\log 11}{\log 2}$ change of base
2. $\boxed{\approx 3.4594}$ calculator

Answer key · solutions	الحلول
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13 · Practice · Ms. Fatima Al Saeedi · slide 29 · answer C

Express in terms of common logarithms, then approximate to the nearest ten-thousandth. $\log_8 30$

1. $\log_8 30 = \dfrac{\log 30}{\log 8}$ change of base
2. ≈ 1.6356 calculator

14 · Practice · Ms. Fatima Al Saeedi · slide 29 · answer B

Express in terms of common logarithms, then approximate to the nearest ten-thousandth. $\log_5 0.4$

1. $\log_5 0.4 = \dfrac{\log 0.4}{\log 5}$ change of base
2. ≈ -0.5693 negative: 0.4 < 1

15 · Practice · Ms. Fatima Al Saeedi · slide 31 · answer A

Write the expression as a single logarithm. $\frac{1}{3}\ln 8 - \ln 3 + \ln 9$

1. $\ln 2 - \ln 3 + \ln 9 = \ln \dfrac{2 \cdot 9}{3} = \ln 6$ use the power rule, then the product / quotient rule

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16 · Practice · Ms. Fatima Al Saeedi · slide 31 · answer D

Write the expression as a single logarithm. $\ln 7 - \frac{1}{3}\ln 27 + \ln 6$

1 $\ln 7 - \ln 3 + \ln 6 = \ln \frac{7 \cdot 6}{3} = \ln 14$ use the power rule, then the product / quotient rule

17 · Practice · Ms. Fatima Al Saeedi · slide 31 · answer B

Write the expression as a single logarithm. $3\ln 3 - \ln 9$

1 $\ln 27 - \ln 9 = \ln \frac{27}{9} = \ln 3$ use the power rule, then the product / quotient rule

18 · Practice · Ms. Fatima Al Saeedi · slide 31 · answer C

Write the expression as a single logarithm. $4\ln 16 - \ln 256$

1 $\ln 16^4 - \ln 256 = \ln \frac{65536}{256} = \ln 256 = 2\ln 16$ use the power rule, then the product / quotient rule

Answer key · solutions	الحلول
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19 · Practice · Ms. Fatima Al Saeedi · slide 31 · answer C

Write the expression as a single logarithm. $2\ln x + 2\ln 4$

1. $\ln x^2 + \ln 16 = \ln(16x^2)$ use the power rule, then the product / quotient rule

20 · Practice · Ms. Fatima Al Saeedi · slide 37 · answer A

Simplify the expression and state when the original expression is undefined. $\dfrac{x(x-3)(x+6)}{x^2+x-12}$

1. $x^2 + x - 12 = (x+4)(x-3)$ factor the denominator
2. $\dfrac{x(x-3)(x+6)}{(x+4)(x-3)} = \dfrac{x(x+6)}{x+4}$ cancel (x − 3)
3. undefined when $(x+4)(x-3) = 0:\ x = -4, 3$ use the ORIGINAL denominator

21 · Practice · Ms. Fatima Al Saeedi · slide 37 · answer C

Simplify the expression and state when the original expression is undefined. $\dfrac{y^2(y^2+3y+2)}{2y(y-4)(y+2)}$

1. $y^2 + 3y + 2 = (y+1)(y+2)$ factor
2. $\dfrac{y^2(y+1)(y+2)}{2y(y-4)(y+2)} = \dfrac{y(y+1)}{2(y-4)}$ cancel y and (y + 2)
3. undefined when $2y(y-4)(y+2) = 0:\ y = 0, 4, -2$ original denominator

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22 · Practice · Ms. Fatima Al Saeedi · slide 40 · answer B

Simplify. $\frac{3}{x}+\frac{5}{y}$

1. $\text{LCD}=xy$
2. $\boxed{\frac{3y}{xy}+\frac{5x}{xy}=\frac{5x+3y}{xy}}$ multiply top and bottom to get the LCD

23 · Practice · Ms. Fatima Al Saeedi · slide 40 · answer C

Simplify. $\frac{3}{8p^2r}+\frac{5}{4p^2r}$

1. $\text{LCD}=8p^2r$
2. $\boxed{\frac{3}{8p^2r}+\frac{10}{8p^2r}=\frac{13}{8p^2r}}$ 5 × 2 = 10

24 · Practice · Ms. Fatima Al Saeedi · slide 40 · answer A

Simplify. $\frac{2c-7}{3}+4$

1. $4=\frac{12}{3}$
2. $\boxed{\frac{2c-7+12}{3}=\frac{2c+5}{3}}$

Answer key · solutions	الحلول
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25 · Practice · Ms. Fatima Al Saeedi · slide 45 · answer C

Find the vertical and oblique asymptotes of $f(x) = \dfrac{(x-4)^2}{x+2}$.

1. $x + 2 = 0 \Rightarrow x = -2$ vertical asymptote
2. $\dfrac{x^2 - 8x + 16}{x+2} = x - 10 + \dfrac{36}{x+2}$ long division
3. oblique: $f(x) = x - 10$

26 · Practice · Ms. Fatima Al Saeedi · slide 47 · answer A

Find the zeros and asymptotes of $f(x) = \dfrac{6x^2 + 4x + 2}{x+2}$. Which pair is correct?

1. $6x^2 + 4x + 2 = 0 : \ \Delta = 16 - 48 < 0$ no real zeros
2. $x = -2$ vertical asymptote
3. $\dfrac{6x^2 + 4x + 2}{x+2} = 6x - 8 + \dfrac{18}{x+2}$ long division
4. oblique: $f(x) = 6x - 8$

27 · Practice · Ms. Fatima Al Saeedi · slide 59 · answer A

Solve the equation. $\dfrac{2x+3}{x+1} = \dfrac{3}{2}$

1. $2(2x + 3) = 3(x + 1)$ cross-multiply
2. $4x + 6 = 3x + 3$
3. $x = -3$ check: x ≠ −1 ✓

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28 · Practice · Ms. Fatima Al Saeedi · slide 59 · answer A

Solve the equation. $\frac{-12}{y} = y - 7$

1. $-12 = y(y-7)$ multiply by y
2. $y^2 - 7y + 12 = 0$
3. $\boxed{(y-3)(y-4) = 0 \Rightarrow y = 3,\ 4}$ both work: y ≠ 0 ✓

29 · Practice · Ms. Fatima Al Saeedi · slide 59 · answer B

Solve the equation. $\frac{9}{x-7} - \frac{7}{x-6} = \frac{13}{x^2 - 13x + 42}$

1. $x^2 - 13x + 42 = (x-7)(x-6)$ LCD
2. $9(x-6) - 7(x-7) = 13$ multiply by the LCD
3. $\boxed{2x - 5 = 13 \Rightarrow x = 9}$ check: x ≠ 6, 7 ✓

30 · Practice · Ms. Fatima Al Saeedi · slide 62 · answer B

Identify the question or sample as biased or unbiased: “Every tenth student at band camp is asked to name his or her favorite band for a survey about the campers.”

1. Unbiased every 10th student is a fair systematic sample

Answer key · solutions	الحلول
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31 · Practice · Ms. Fatima Al Saeedi · slide 62 · answer A

Identify the question or sample as biased or unbiased: “Which is your favorite type of music, pop or country?”

1 Biased it offers only two choices, so it leads the answer

32 · Practice · Ms. Fatima Al Saeedi · slide 62 · answer A

Identify the question or sample as biased or unbiased: “Don’t you agree that employees should pack their lunch?”

1 Biased the wording pushes toward “yes”

33 · Practice · Ms. Fatima Al Saeedi · slide 62 · answer B

Identify the question or sample as biased or unbiased: “Are you a member of any after-school clubs?”

1 Unbiased clear wording, all answers possible

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34 · Practice · Ms. Fatima Al Saeedi · slide 63 · answer A

A spinner with 4 equal sections is spun 100 times: Red 35, Blue 38, Green 13, Yellow 14. Find the THEORETICAL probability of blue.

1. 4 equal sections
2. $\boxed{P(\text{blue}) = \frac{1}{4} = 25\%}$ theory ignores the results

35 · Practice · Ms. Fatima Al Saeedi · slide 63 · answer D

Same spinner (Red 35, Blue 38, Green 13, Yellow 14 in 100 spins). Find the EXPERIMENTAL probability of blue.

1. $P(\text{blue}) = \frac{38}{100}$ use the results
2. $\boxed{= 38\%}$

36 · Practice · Ms. Fatima Al Saeedi · slide 64 · answer B

A coin is flipped 125 times: 73 heads, 52 tails. Find the THEORETICAL probability of heads.

1. 2 equal outcomes
2. $\boxed{P(H) = \frac{1}{2} = 50\%}$

Answer key · solutions	الحلول
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37 · Practice · Ms. Fatima Al Saeedi · slide 64 · answer C

A coin is flipped 125 times: 73 heads, 52 tails. Find the EXPERIMENTAL probability of heads.

1. $P(H) = \frac{73}{125}$
2. $\boxed{= 58.4\%}$

38 · Practice · Ms. Fatima Al Saeedi · slide 65 · answer C

A fair die is rolled 150 times and a 3 appears 27 times. Find the EXPERIMENTAL probability of rolling a 3.

1. $P(3) = \frac{27}{150}$
2. $\boxed{= 18\%}$

39 · Practice · Ms. Fatima Al Saeedi · slide 66 · answer B

A card is drawn from a standard deck and returned, 120 times: Heart 28, Diamond 37, Spade 34, Club 21. Which statement is true?

1. theoretical $P = \frac{13}{52} = 25\%$ for every suit
2. diamond: $\frac{37}{120} \approx 30.8\% > 25\%$ so the statement is true
3. $\boxed{\text{heart } 23.3\% < 25\%,\ \text{club } 17.5\% < 25\%}$ the others are false

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40 · Practice · Ms. Fatima Al Saeedi · slide 68 · answer C

Race times (s): 57.1, 55.9, 59.3, 54.9, 54.6, 50.3, 55.2, 53.5. Find the standard deviation of the data.

1. $\mu = \dfrac{440.8}{8} = 55.1$
2. $\boxed{\sigma = \sqrt{\dfrac{\sum(x-\mu)^2}{8}} \approx 2.45}$ population standard deviation

41 · Practice · Ms. Fatima Al Saeedi · slide 68 · answer B

Prices (AED): 50, 165, 55, 79, 84, 68, 38, 42. Find the standard deviation of the data.

1. $\mu = \dfrac{581}{8} = 72.625$
2. $\boxed{\sigma = \sqrt{\dfrac{\sum(x-\mu)^2}{8}} \approx 38.22}$

42 · Practice · Ms. Fatima Al Saeedi · slide 70 · answer B

Rewrite the measure in the other unit. 18°

1. $\boxed{18 \cdot \dfrac{\pi}{180} = \dfrac{\pi}{10}}$ multiply by the conversion factor

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43 · Practice · Ms. Fatima Al Saeedi · slide 70 · answer A

Rewrite the measure in the other unit. $-72°$

1 $-72 \cdot \frac{\pi}{180} = -\frac{2\pi}{5}$ multiply by the conversion factor

44 · Practice · Ms. Fatima Al Saeedi · slide 70 · answer B

Rewrite the measure in the other unit. $\frac{5\pi}{2}$

1 $\frac{5\pi}{2} \cdot \frac{180}{\pi} = 450°$ multiply by the conversion factor

45 · Practice · Ms. Fatima Al Saeedi · slide 70 · answer D

Rewrite the measure in the other unit. $-\frac{7\pi}{12}$

1 $-\frac{7\pi}{12} \cdot \frac{180}{\pi} = -105°$ multiply by the conversion factor

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46 · Practice · Ms. Fatima Al Saeedi · slide 71 · answer B

Find the exact value. $\tan 330^\circ$

1. 330° is in QIV, reference 30° : $\tan = -\tan 30^\circ = -\dfrac{\sqrt{3}}{3}$

47 · Practice · Ms. Fatima Al Saeedi · slide 71 · answer C

Find the exact value. $\cos\left(-\dfrac{11\pi}{4}\right)$

1. $-\dfrac{11\pi}{4} + 4\pi = \dfrac{5\pi}{4}$ (QIII, reference $\frac{\pi}{4}$) : $\cos = -\dfrac{\sqrt{2}}{2}$

48 · Practice · Ms. Fatima Al Saeedi · slide 71 · answer D

Find the exact value. $\cot 30^\circ$

1. $\cot 30^\circ = \dfrac{1}{\tan 30^\circ} = \sqrt{3}$

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49 · Practice · Ms. Fatima Al Saeedi · slide 71 · answer A

Find the exact value. $\sin(-150°)$

1 $-150°$ is in QIII, reference $30°$: $\sin = -\frac{1}{2}$

50 · Practice · Ms. Fatima Al Saeedi · slide 72 · answer C

The terminal side of angle θ in standard position meets the unit circle at $P\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$. Find $\sin\theta$ and $\cos\theta$.

1 $P(\cos\theta, \sin\theta) \Rightarrow \cos\theta = -\frac{\sqrt{3}}{2},\ \sin\theta = \frac{1}{2}$ x = cos θ, y = sin θ

51 · Practice · Ms. Fatima Al Saeedi · slide 72 · answer A

The terminal side of angle θ in standard position meets the unit circle at $P(0, -1)$. Find $\sin\theta$ and $\cos\theta$.

1 $\cos\theta = 0,\ \sin\theta = -1$ x = cos θ, y = sin θ

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52 · Practice · Ms. Fatima Al Saeedi · slide 72 · answer D

The terminal side of angle θ in standard position meets the unit circle at $P\left(-\frac{2}{3}, \frac{\sqrt{5}}{3}\right)$. Find $\sin\theta$ and $\cos\theta$.

1. $\cos\theta = -\frac{2}{3},\ \sin\theta = \frac{\sqrt{5}}{3}$ x = cos θ, y = sin θ

53 · Practice · Ms. Fatima Al Saeedi · slide 74 · answer B

A spring oscillates by $y = 40\cos\pi t$ (y in cm, t in seconds). Find the period.

1. $\text{period} = \frac{2\pi}{|b|} = \frac{2\pi}{\pi} = 2$ b = π

54 · Practice · Ms. Fatima Al Saeedi · slide 74 · answer C

For $y = 40\cos\pi t$, find the range.

1. $\text{amplitude} = 40,\ \text{midline}\ y = 0$
2. $[-40, 40]$

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55 · Practice · Ms. Fatima Al Saeedi · slide 83 · answer C

For $y = \cos\theta + 3$, what are the amplitude and the midline?

1. $y = a\cos b(\theta - h) + k:\ a = 1,\ k = 3$
2. $\boxed{\text{midline } y = k = 3}$

56 · Practice · Ms. Fatima Al Saeedi · slide 83 · answer C

For $y = \cos\theta + 3$, find the range.

1. $-1 \le \cos\theta \le 1$
2. $\boxed{2 \le \cos\theta + 3 \le 4}$ add 3

57 · Practice · Ms. Fatima Al Saeedi · slide 87 · answer A

For $y = 2\sin\theta - 4$, which statement is correct?

1. $a = 2,\ b = 1 \Rightarrow \text{period } \dfrac{360^\circ}{1} = 360^\circ$
2. $\boxed{k = -4 \Rightarrow \text{midline } y = -4\ (4 \text{ down})}$

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58 · Practice · Ms. Fatima Al Saeedi · slide 87 · answer B

For $y = 2\sin\theta - 4$, find the range.

1. $-1 \le \sin\theta \le 1 \Rightarrow -2 \le 2\sin\theta \le 2$
2. $\boxed{-6 \le 2\sin\theta - 4 \le -2}$ subtract 4

59 · Practice · Ms. Fatima Al Saeedi · slide 96 · answer A

Find the value in degrees. $\tan^{-1}\sqrt{3}$

1. $\tan 60° = \sqrt{3}$ range of $\tan^{-1}$: (−90°, 90°)
2. $\boxed{60°}$

60 · Practice · Ms. Fatima Al Saeedi · slide 96 · answer A

Find the value in radians. $\cos^{-1}\left(\frac{\sqrt{2}}{2}\right)$

1. $\cos\frac{\pi}{4} = \frac{\sqrt{2}}{2}$ range of $\cos^{-1}$: [0, π]
2. $\boxed{\frac{\pi}{4}}$

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61 · Practice · Ms. Fatima Al Saeedi · slide 19 · answer B

Which description matches $f(x) = -2e^{x-3} + 2$?

1. $f(x) = a\,e^{x-h} + k$ compare
2. $a = -2,\ h = 3,\ k = 2$
3. $\boxed{|a| = 2 > 1 \Rightarrow \text{stretch}}$

62 · Practice · Ms. Fatima Al Saeedi · slide 34 · answer C

An experiment starts with 20 bacteria A cells. After 45 minutes there are 710. Which equation models the number of cells y after t minutes? ($y = ae^{kt}$)

1. $a = 20$ initial population
2. $710 = 20e^{45k} \Rightarrow k = \dfrac{\ln 35.5}{45} \approx 0.079$
3. $\boxed{y = 20e^{0.079t}}$

63 · Practice · Ms. Fatima Al Saeedi · slide 34 · answer D

Using $y = 20e^{0.079t}$, after how many minutes will there be 1000 bacteria A cells?

1. $1000 = 20e^{0.079t}$
2. $50 = e^{0.079t}$
3. $\boxed{t = \dfrac{\ln 50}{0.079} \approx 49.5}$ take ln of both sides

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64 · Practice · Ms. Fatima Al Saeedi · slide 35 · answer B

In 2000 the world population was 6.124 billion; in 2005 it was 6.515 billion. Which equation models the population y (billions) t years after 2000?

1. $a = 6.124$
2. $6.515 = 6.124e^{5k} \Rightarrow k = \dfrac{\ln(6.515/6.124)}{5} \approx 0.0124$
3. $\boxed{y = 6.124e^{0.0124t}}$

65 · Practice · Ms. Fatima Al Saeedi · slide 35 · answer B

Using $y = 6.124e^{0.0124t}$, in which year does the world population reach 7.5 billion?

1. $7.5 = 6.124e^{0.0124t}$
2. $t = \dfrac{\ln(7.5/6.124)}{0.0124} \approx 16.3$
3. $\boxed{2000 + 16.3 \approx 2016}$

66 · Practice · Ms. Fatima Al Saeedi · slide 106 · answer A

The graph of $g(x) = \dfrac{a}{x-h} + k$ has vertical asymptote x = −4, horizontal asymptote y = 0, passing through (5, 1). Which function is it?

1. $x = h,\ y = k$ are the asymptotes
2. $h = -4,\ k = 0:\ 1 = \dfrac{a}{5+4} \Rightarrow a = 9$ substitute the given point
3. $\boxed{g(x) = \dfrac{9}{x+4}}$

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67 · Practice · Ms. Fatima Al Saeedi · slide 107 · answer B

The graph of $g(x) = \dfrac{a}{x-h} + k$ has vertical asymptote x = −4, horizontal asymptote y = −2, passing through (−5, 4). Which function is it?

1. $x = h,\ y = k$ are the asymptotes
2. $h = -4,\ k = -2:\ 4 = \dfrac{a}{-5+4} - 2 \Rightarrow -a = 6 \Rightarrow a = -6$ substitute the given point
3. $\boxed{g(x) = \dfrac{-6}{x+4} - 2}$

68 · Practice · Ms. Fatima Al Saeedi · slide 107 · answer B

The graph of $g(x) = \dfrac{a}{x-h} + k$ has vertical asymptote x = 2, horizontal asymptote y = 2, passing through (7, 1). Which function is it?

1. $x = h,\ y = k$ are the asymptotes
2. $h = 2,\ k = 2:\ 1 = \dfrac{a}{7-2} + 2 \Rightarrow a = -5$ substitute the given point
3. $\boxed{g(x) = \dfrac{-5}{x-2} + 2}$

69 · Practice · Ms. Fatima Al Saeedi · slide 109 · answer B

Find the z-value. $X = 24,\ \mu = 19,\ \sigma = 3.8$

1. $z = \dfrac{X - \mu}{\sigma}$
2. $\boxed{z = \dfrac{24 - 19}{3.8} \approx 1.32}$ to the nearest hundredth

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70 · Practice · Ms. Fatima Al Saeedi · slide 109 · answer C

Find the z-value. $X = 106.3,\ \mu = 88.8,\ \sigma = 9.6$

1. $z = \dfrac{X - \mu}{\sigma}$
2. $\boxed{z = \dfrac{106.3 - 88.8}{9.6} \approx 1.82}$ to the nearest hundredth

71 · Practice · Ms. Fatima Al Saeedi · slide 109 · answer D

Find the z-value. $\sigma = 9.8,\ X = 55.4,\ \mu = 68.34$

1. $z = \dfrac{X - \mu}{\sigma}$
2. $\boxed{z = \dfrac{55.4 - 68.34}{9.8} \approx -1.32}$ to the nearest hundredth

72 · Practice · Ms. Fatima Al Saeedi · slide 120 · answer B

Find the period of $y = \tan 5x$.

1. $\boxed{\text{period} = \dfrac{180^\circ}{|b|} = \dfrac{180^\circ}{5} = 36^\circ}$

Answer key · solutions	الحلول
Answer key · solutions	مفتاح الإجابة والحلول

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Find the asymptotes of $y = \tan 5x$ (n is an integer).

1. $x = \dfrac{(90 + 180n)^\circ}{|b|} = \dfrac{(90 + 180n)^\circ}{5}$

2. $\boxed{x = (18 + 36n)^\circ}$